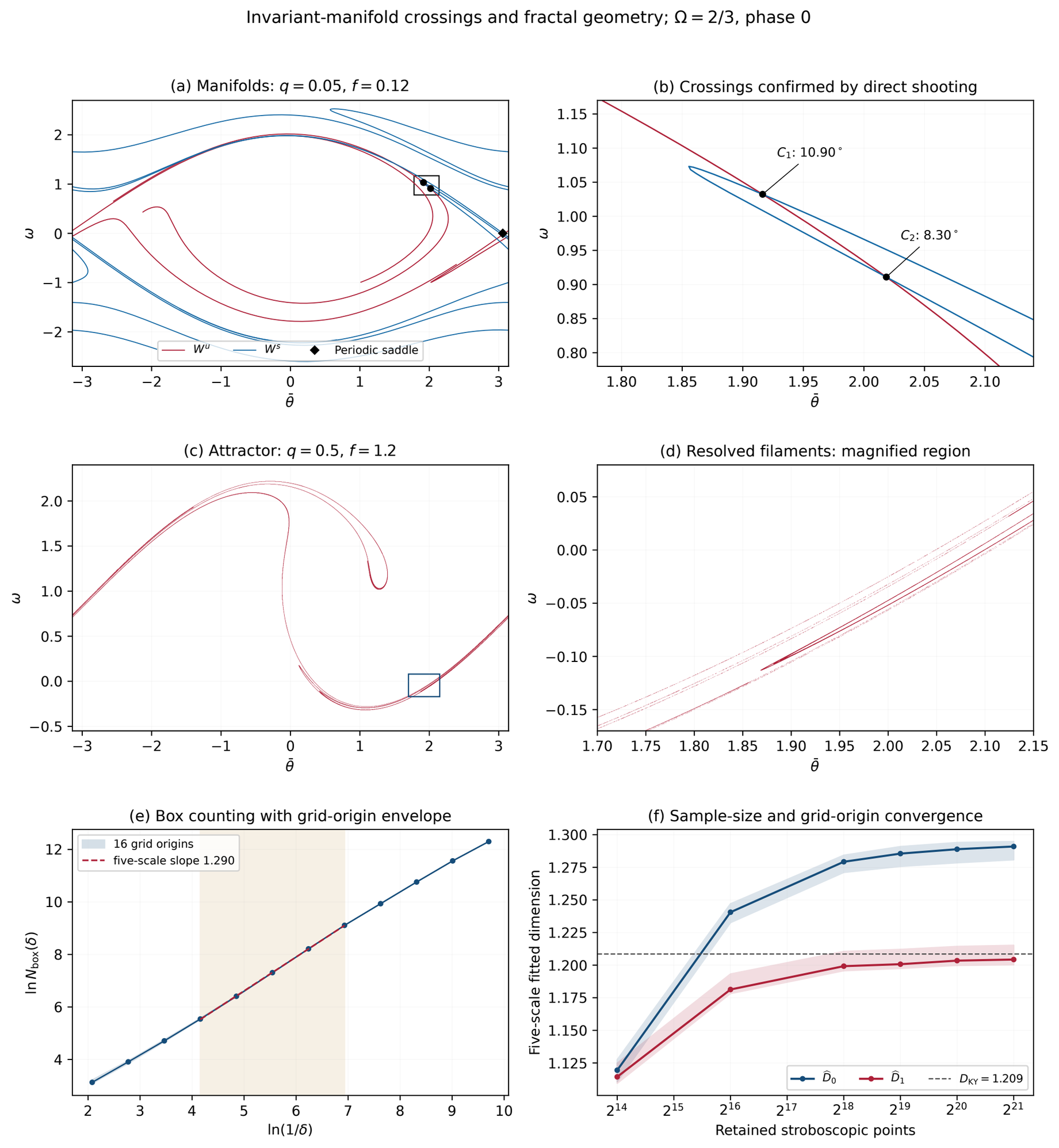

Invariant-manifold crossings and fractal geometry; Ω = 2/3, phase 0
(a) Manifolds: q = 0.05, f = 0.12
W^u
W^s
Periodic saddle
ω
θ̄
(b) Crossings confirmed by direct shooting
C1: 10.90°
C2: 8.30°
ω
θ̄
(c) Attractor: q = 0.5, f = 1.2
ω
θ̄
(d) Resolved filaments: magnified region
ω
θ̄
(e) Box counting with grid-origin envelope
16 grid origins
five-scale slope 1.290
ln N_box(δ)
ln(1/δ)
(f) Sample-size and grid-origin convergence
D̂0
D̂1
D_KY = 1.209
Five-scale fitted dimension
Retained stroboscopic points